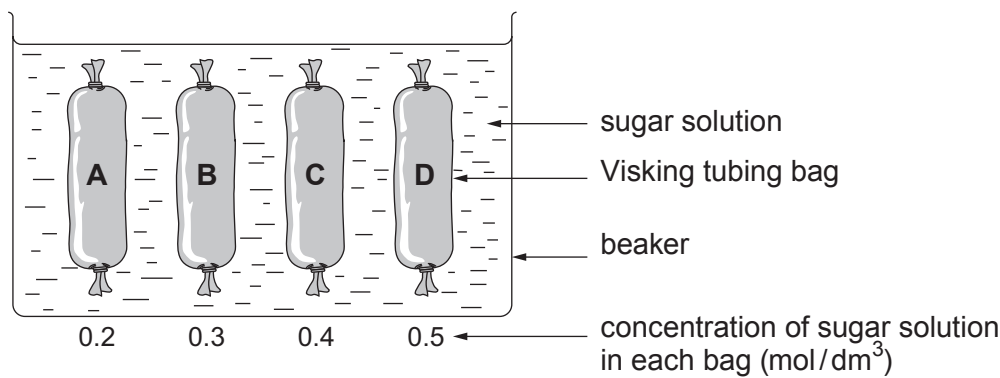


4. A group of students carried out the following investigation.
- Four identical Visking tubing bags were filled with 10 cm³ of a sugar solution.
 - Each bag contained a different concentration of sugar solution.
 - The outsides of the bags were washed and dried and then the mass of each bag was recorded.
 - The bags were placed in a large beaker containing a sugar solution of unknown concentration.
 - The bags were left for 30 minutes.
 - The bags were removed from the beaker, the outsides of the bags were dried and their final mass recorded.
 - The percentage change in mass was calculated.



The results are shown in the table.

Visking tubing bag	Concentration of sugar in the Visking tubing bag (mol/dm ³)	Percentage change in mass at 30 minutes (%)
A	0.2	-2.6
B	0.3	+1.3
C	0.4	+3.6
D	0.5	+4.2



Examiner
only

(a) Explain the result for Visking tubing bag **D**.

[4]

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(b) Estimate the concentration of the sugar solution in the beaker. Give a reason for your answer. [2]

Estimated concentration = mol/dm³

Reason:

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